

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow engine** for MCA company-registration filings, reducing manual filing time by **85%** through state recovery, retries, and public-site API lookups.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized services with **Docker** and automated **GitHub Actions** deployments to **Google Cloud Run**, improving release velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built MRI segmentation tooling for **95 anatomical regions**, turning model outputs into usable masks and 3D visualization artifacts inside an existing application workflow.
- Reduced processing runtime by **87%** with async/parallel execution and fewer repeated post-processing passes over large MRI volumes.
- Integrated outputs into 3D reconstructed surface visualization flows so model results stayed inspectable and useful to end users.

Open Source

[Contributions](#) 

- Contributed to Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working with existing maintainers, code style, and review constraints.

Projects

Swish Support System | *Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse*

[GitHub](#) 

- Built AI-assisted backend workflows for order status, delays, refunds, trust checks, and escalation using tool calls over operational APIs with PostgreSQL and Redis state.
- Added retrieval-backed responses, confidence thresholds, human handoff paths, conversation memory, and traceable action records for ambiguous or high-risk user workflows.

Distributed Inference Engine | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

Seaweed: Contest Platform | *Go, React, PostgreSQL, Judge0, Redis, AWS, Docker, Kubernetes*

[GitHub](#) 

- Architected a full-stack contest platform supporting **500+ concurrent users**, automated judging, live rankings, and feedback loops for programming contest workflows.
- Designed a sandboxed Judge0 pipeline with async Go dispatch, Docker/Kubernetes deployment on AWS, **sub-5s P95 verdict latency**, per-test telemetry, and PostgreSQL-backed result analysis.

Technical Skills

Languages: Python, Go, TypeScript

Frameworks: FastAPI, Gin, Echo, Node.js, Agno

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Langfuse, OpenTelemetry, GitHub Actions, AWS